

Digital SAT Math

Topic 1: Quadratic Equations - Solving

Complete Solutions & Answer Key

Total: 60 Questions

Format: Digital SAT Style — MCQ & Grid-In

Difficulty: Easy (Q1-20) • Medium (Q21-45) • Hard (Q46-60)

Q1. Correct Answer: C

STEP-BY-STEP SOLUTION:

$(x-2)(x-7)=0 \rightarrow x=2$ or $x=7$. Answer C is $x=2$.

WHY EACH OPTION IS WRONG:

A) -7: $(-7)^2-9(-7)+14=126 \neq 0$

B) -2: $4+18+14=36 \neq 0$

D) 9: $81-81+14=14 \neq 0$

TRAP ALERT: Roots are +2 and +7, not -2 and -7.

KEY INSIGHT: Product=14, Sum=9 $\rightarrow$ roots are 2 and 7.

FAST METHOD: $2 \times 7 = 14$, $2 + 7 = 9$ ✓

Q2. Correct Answer: 4

STEP-BY-STEP SOLUTION:

$x^2=16 \rightarrow x=\pm 4$. Positive=4

TRAP ALERT: Forgetting $\pm$ when taking square root.

KEY INSIGHT: Difference of squares: $x^2-16=(x+4)(x-4)$.

FAST METHOD: $\sqrt{16}=4$

Q3. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$(x+5)^2 = x^2 + 10x + 25 \checkmark$$

WHY EACH OPTION IS WRONG:

B) $(x-5)^2 = x^2 - 10x + 25$ (wrong sign)

C) $(x+5)(x-5) = x^2 - 25$ (diff of squares)

D) $(x+25)(x+1) = x^2 + 26x + 25$

TRAP ALERT: Confusing perfect square with difference of squares.

KEY INSIGHT: $2(x)(5) = 10x$ confirms perfect square.

FAST METHOD: Middle term $10x = 2(x)(5) \rightarrow (x+5)^2$.

Q4. Correct Answer: 6

STEP-BY-STEP SOLUTION:

$$x(x-6)=0 \rightarrow x=0 \text{ or } x=6. \text{ Positive}=6$$

TRAP ALERT: Missing $x=0$ as the other solution.

KEY INSIGHT: Always factor out GCF (x) first.

FAST METHOD: $x(x-6)=0 \rightarrow x=6$.

Q5. Correct Answer: B

STEP-BY-STEP SOLUTION:

$$x^2 + 4 = 0 \rightarrow x^2 = -4. \text{ No real number squared is negative. No real solutions.}$$

WHY EACH OPTION IS WRONG:

A) $x^2 - 4 = 0 \rightarrow x = \pm 2$

C) $x^2 - 4x = 0 \rightarrow x = 0 \text{ or } 4$

D) $\text{disc} = 16 + 20 = 36 > 0 \rightarrow \text{two solutions}$

TRAP ALERT: Confusing 'no solutions' with 'one solution.'

KEY INSIGHT: $x^2 = (\text{negative})$ has no real solutions.

FAST METHOD: $x^2 = -4$ impossible for real x.

Q6. Correct Answer: C

STEP-BY-STEP SOLUTION:

$$\text{Area} = w \times (w+3) = 40$$

WHY EACH OPTION IS WRONG:

- A) $w^2+3=40$ drops multiplication
- B) $2w+3=40$ is perimeter-type
- D) $w+(w+3)=40$ adds instead of multiplying

TRAP ALERT: Mixing perimeter (add) with area (multiply).

KEY INSIGHT: Area = length $\times$ width.

FAST METHOD: $w(w+3)=40$ directly.

Q7. Correct Answer: 2

STEP-BY-STEP SOLUTION:

$$2x^2=8 \rightarrow x^2=4 \rightarrow x=2 \text{ (positive)}$$

TRAP ALERT: Not dividing by 2 first, getting $x^2=8$.

KEY INSIGHT: Isolate x^2 before square rooting.

FAST METHOD: $x^2=4 \rightarrow x=2$.

Q8. Correct Answer: B

STEP-BY-STEP SOLUTION:

$$x^2-7x+12=(x-3)(x-4). \text{ Factor } (x-3) \text{ is in the choices.}$$

WHY EACH OPTION IS WRONG:

- A) $f(2)=4-14+12=2 \neq 0$
- C) $(x+4)$ gives wrong product sign
- D) $(x+6)$ is not a factor

TRAP ALERT: Both roots negative confusion: roots are +3 and +4.

KEY INSIGHT: Product=12, Sum=7 $\rightarrow$ 3 and 4.

FAST METHOD: $3 \times 4=12$, $3+4=7 \rightarrow (x-3)(x-4)$.

Q9. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$\text{Sub } x=2: 4+2b+18=0 \rightarrow 2b=-22 \rightarrow b=-11$$

WHY EACH OPTION IS WRONG:

B) $b=-9: 4-18+18=4 \neq 0$

C) $b=9: 40 \neq 0$

D) $b=11: 44 \neq 0$

TRAP ALERT: Substituting but not setting equal to zero.

KEY INSIGHT: If r is a root: $f(r)=0$. Substitute and solve.

FAST METHOD: $4+2b+18=0 \rightarrow b=-11$.

Q10. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$(x+7)(x-3)=x^2-3x+7x-21=x^2+4x-21$$

WHY EACH OPTION IS WRONG:

B) $x^2-4x-21=(x-7)(x+3)$

C) Last term must be -21

D) $x^2-10x+21=(x-7)(x-3)$

TRAP ALERT: Middle term: $-3+7=+4$, not -4.

KEY INSIGHT: FOIL: outer+inner= $-3x+7x=+4x$.

FAST METHOD: Sum= $+4$ (middle), product= -21 (last).

Q11. Correct Answer: D

STEP-BY-STEP SOLUTION:

$$(x-1)(x-3)=0 \rightarrow x=1 \text{ or } x=3. \text{ Sum}=4$$

WHY EACH OPTION IS WRONG:

A) $-b/a=(-4)/1=+4$, not -4

B) 1 is one root

C) 3 is the other root

TRAP ALERT: Sum= $-b/a=+4$, students think it's -4.

KEY INSIGHT: Sum of roots = $-b$ for monic quadratic.

FAST METHOD: Sum= $-b/a=4$.

Q12. Correct Answer: -6

STEP-BY-STEP SOLUTION:

$$(x+6)^2=0 \rightarrow x=-6$$

TRAP ALERT: Expecting two distinct solutions from a perfect square.

KEY INSIGHT: Double root: parabola tangent to x-axis.

FAST METHOD: $\sqrt{36}=6 \rightarrow x=-6$.

Q13. Correct Answer: C

STEP-BY-STEP SOLUTION:

$$\text{disc}=36-36=0 \rightarrow \text{exactly one real solution.}$$

WHY EACH OPTION IS WRONG:

A) $\text{disc}=36>0$, two solutions

B) $\text{disc}<0$, no solutions

D) $\text{disc}=4>0$, two distinct solutions

TRAP ALERT: Picking B (no solutions) when asked for exactly one.

KEY INSIGHT: $\text{disc}=0 \rightarrow$ one repeated root.

FAST METHOD: $36-36=0 \rightarrow$ one solution.

Q14. Correct Answer: 5

STEP-BY-STEP SOLUTION:

$$3x^2=75 \rightarrow x^2=25 \rightarrow x=5 \text{ (positive)}$$

TRAP ALERT: Forgetting to divide by 3 first.

KEY INSIGHT: Isolate x^2 first.

FAST METHOD: $x^2=25 \rightarrow x=5$.

Q15. Correct Answer: B

STEP-BY-STEP SOLUTION:

$$(x-2)(x-3)=0 \rightarrow x=2 \text{ or } x=3. \text{ Since } x>2: x=3$$

WHY EACH OPTION IS WRONG:

A) $x=2$ satisfies $f(x)=0$ but NOT $x>2$ (strict inequality)

C) $x=5$: $25-25+6=6 \neq 0$

D) $x=6$: $36-30+6=12 \neq 0$

TRAP ALERT: $x=2$ is a root but fails the strict condition $x>2$.

KEY INSIGHT: Find all roots first, then apply stated conditions.

FAST METHOD: Roots 2 and 3; $x>2 \rightarrow x=3$.

Q16. Correct Answer: A**STEP-BY-STEP SOLUTION:**

$(x-4)(x+9)=0 \rightarrow x=4$ or $x=-9$. Answer A is -9.

WHY EACH OPTION IS WRONG:

- B) -4 comes from $(x+4)=0$, not $(x-4)=0$
- C) 5 not a root
- D) 13 not a root

TRAP ALERT: Sign flip: $x+9=0$ gives $x=-9$, not +9.

KEY INSIGHT: Zero Product: each factor set to 0. Sign flips.

FAST METHOD: $x+9=0 \rightarrow x=-9$.

Q17. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$-16t^2+32t=0 \rightarrow -16t(t-2)=0 \rightarrow t=0$ or $t=2$. Positive $t=2$

WHY EACH OPTION IS WRONG:

- A) $t=1$: $h=16$ ft (still airborne)
- C) $t=4$: $h<0$ (invalid)
- D) $t=16$ nonsensical

TRAP ALERT: $t=0$ is also a root (launch); take the positive landing time.

KEY INSIGHT: Factor out $-16t$: roots are 0 (launch) and 2 (landing).

FAST METHOD: $-16t(t-2)=0 \rightarrow t=2$.

Q18. Correct Answer: C**STEP-BY-STEP SOLUTION:**

$x^2-49=x^2-7^2=(x+7)(x-7)$

WHY EACH OPTION IS WRONG:

- A) $(x-7)^2=x^2-14x+49$
- B) $(x+7)^2=x^2+14x+49$
- D) $(x-49)(x+1)=x^2-48x-49$

TRAP ALERT: Difference of squares has NO middle term.

KEY INSIGHT: $a^2-b^2=(a+b)(a-b)$.

FAST METHOD: $a=x$, $b=7 \rightarrow (x+7)(x-7)$.

Q19. Correct Answer: A**STEP-BY-STEP SOLUTION:**

Roots: 3 and -5. $\text{Sum}=3+(-5)=-2=-m \rightarrow m=+2$. $\text{Product}=3 \times (-5)=-15=n$. Equation: $x^2+2x-15=0$

WHY EACH OPTION IS WRONG:

- B) roots 5 and -3 (wrong)
- C) roots -3 and -5 (wrong)
- D) roots 3 and 5 (wrong)

TRAP ALERT: $\text{Sum}=-2 \rightarrow$ coefficient of x is $+2$ (not -2). $b=-\text{sum}$.

KEY INSIGHT: Vieta's: $\text{sum}=-b$, $\text{product}=c$ for x^2+bx+c .

FAST METHOD: $\text{Sum}=-2 \rightarrow b=+2$; $\text{product}=-15 \rightarrow c=-15$.

Q20. Correct Answer: 7**STEP-BY-STEP SOLUTION:**

$(x-4)(x-7)=0 \rightarrow x=4$ or $x=7$. $x>5 \rightarrow x=7$

TRAP ALERT: Both 4 and 7 are roots; condition $x>5$ eliminates $x=4$.

KEY INSIGHT: Find all roots, then apply the condition.

FAST METHOD: Roots 4 and 7; $x>5 \rightarrow 7$.

Q21. Correct Answer: B**STEP-BY-STEP SOLUTION:**

Let $u=2x-1$. $u^2-9u+20=0 \rightarrow (u-4)(u-5)=0 \rightarrow u=4$ or $u=5$.

$u=4$: $2x-1=4 \rightarrow x=5/2$. $u=5$: $2x-1=5 \rightarrow x=3$.

Smaller solution = $5/2$

WHY EACH OPTION IS WRONG:

- A) $3/2$: gives $u=2$, not a root
- C) $x=3$ is the LARGER solution
- D) $7/2$: gives $u=6$, not a root

TRAP ALERT: Forgetting to convert back from u to x .

KEY INSIGHT: Substitution method: solve for u , then back-solve for x .

FAST METHOD: $u=4 \rightarrow x=5/2$ (smaller); $u=5 \rightarrow x=3$.

Q22. Correct Answer: B

STEP-BY-STEP SOLUTION:

Sub $(-3,0)$: $9-3k-12=0 \rightarrow -3-3k=0 \rightarrow k=-1$

WHY EACH OPTION IS WRONG:

A) $k=-7$: $9+21-12=18 \neq 0$

C) $k=1$: $9-3-12=-6 \neq 0$

D) $k=7$: $9-21-12=-24 \neq 0$

TRAP ALERT: Plugging in but not setting equal to zero.

KEY INSIGHT: Graph passes through (a,b) means $f(a)=b$.

FAST METHOD: $9-3k-12=0 \rightarrow k=-1$.

Q23. Correct Answer: C

STEP-BY-STEP SOLUTION:

$4x^2-12x+9=(2x-3)^2=0 \rightarrow x=3/2$

WHY EACH OPTION IS WRONG:

A) $x=1/2$: $1-6+9=4 \neq 0$

B) $x=3/4$: $9/4-9+9=9/4 \neq 0$

D) $x=3$: $36-36+9=9 \neq 0$

TRAP ALERT: Not recognizing perfect square; using quadratic formula unnecessarily.

KEY INSIGHT: $\text{disc}=144-144=0 \rightarrow$ perfect square trinomial.

FAST METHOD: $(2x-3)^2=0 \rightarrow x=3/2$.

Q24. Correct Answer: B

STEP-BY-STEP SOLUTION:

Vertex at $t=-b/(2a)=-30/(2 \times -5)=3$. $h(3)=-5(9)+90+8=-45+90+8=53$

WHY EACH OPTION IS WRONG:

A) $38=h$ at $t=2$

C) 83: arithmetic error

D) 133: far too large

TRAP ALERT: Using wrong time; max is at vertex $t=-b/(2a)$.

KEY INSIGHT: Maximum of downward parabola at $t=-b/(2a)$.

FAST METHOD: $t=3 \rightarrow h=-45+90+8=53$.

Q25. Correct Answer: A

STEP-BY-STEP SOLUTION:

Test $(2x+3)(3x-1)$: outer+inner= $-2x+9x=+7x$ ✓; product= $6x^2+7x-3$ ✓

WHY EACH OPTION IS WRONG:

B) $(6x-1)(x+3)=6x^2+18x-x-3=6x^2+17x-3\neq$

C) $(2x-1)(3x+3)=6x^2+6x-3x-3=6x^2+3x-3\neq$

D) $(3x+1)(2x-3)=6x^2-9x+2x-3=6x^2-7x-3\neq$

TRAP ALERT: Guessing factors without verifying the middle term.

KEY INSIGHT: AC method: product= $6\times(-3)=-18$; find two numbers summing to 7: +9 and -2.

FAST METHOD: $(2x+3)(3x-1)$: check outer+inner= $-2x+9x=+7x$ ✓

Q26. Correct Answer: A

STEP-BY-STEP SOLUTION:

disc= $(-4)^2-4(1)(c)=16-4c>0 \rightarrow c<4$. Only $c=3$ satisfies $c<4$.

WHY EACH OPTION IS WRONG:

B) $c=4$: disc=0, exactly ONE solution

C) $c=5$: disc= $-4<0$, no solutions

D) $c=6$: disc= $-8<0$, no solutions

TRAP ALERT: $c=4$ gives one solution (disc=0), not two distinct ones.

KEY INSIGHT: Two DISTINCT solutions require disc STRICTLY > 0 .

FAST METHOD: $16-4c>0 \rightarrow c<4$. Only A ($c=3$) qualifies.

Q27. Correct Answer: B

STEP-BY-STEP SOLUTION:

$-x^2+14x-40=0 \rightarrow x^2-14x+40=0 \rightarrow (x-4)(x-10)=0 \rightarrow x=4$ or $x=10$

WHY EACH OPTION IS WRONG:

A) $x=2$ and $x=20$: $4-40+40=4\neq 0$

C) $x=5$ and $x=8$: $25-70+40=-5\neq 0$

D) $x=7$ (double): $49-98+40=-9\neq 0$

TRAP ALERT: Forgetting to multiply by -1 when leading coefficient is negative.

KEY INSIGHT: Multiply by -1 first: $x^2-14x+40=0$.

FAST METHOD: $(x-4)(x-10)=0 \rightarrow$ roots 4 and 10.

Q28. Correct Answer: B

STEP-BY-STEP SOLUTION:

$$x^2+8x+15=0 \rightarrow (x+3)(x+5)=0 \rightarrow x=-3 \text{ or } x=-5. \text{ Sum}=-8$$

WHY EACH OPTION IS WRONG:

- A) -15: product of roots
- C) 8: not a root sum
- D) 15: absolute value of product

TRAP ALERT: Not moving -8 to left side first.

KEY INSIGHT: Rearrange to standard form, then apply Vieta's.

FAST METHOD: After simplifying: $\text{sum}=-b/a=-8$.

Q29. Correct Answer: 3

STEP-BY-STEP SOLUTION:

$$f(x)=-2(x-3)^2+18. \text{ Vertex form: } h=3.$$

TRAP ALERT: Reading $k=18$ (y-value) instead of $h=3$ (x-value).

KEY INSIGHT: Vertex form $y=a(x-h)^2+k$. Vertex is (h,k) . $(x-3)$ means $h=+3$.

FAST METHOD: Vertex x-coord=3 directly.

Q30. Correct Answer: 1

STEP-BY-STEP SOLUTION:

$$p+q=5 \text{ and } pq=6 \rightarrow \text{roots of } t^2-5t+6=0 \rightarrow t=2 \text{ or } 3. p=3>q=2 \rightarrow p-q=1$$

TRAP ALERT: Finding $p+q=5$ and stopping.

KEY INSIGHT: Use Vieta's for sum/product; identify which is larger.

FAST METHOD: Roots 3 and 2; $p=3, q=2 \rightarrow p-q=1$.

Q31. Correct Answer: C

STEP-BY-STEP SOLUTION:

$$x^2-6x+8=0 \rightarrow \text{disc}=36-32=4>0 \rightarrow 2 \text{ distinct solutions}$$

WHY EACH OPTION IS WRONG:

- A) $\text{disc}>0$, solutions DO exist
- B) $\text{disc}=0$ needed for 1 solution
- D) impossible for quadratic

TRAP ALERT: Solving $g(x)=0$ instead of $g(x)=-3$.

KEY INSIGHT: Set equal to given value, rearrange, check discriminant.

FAST METHOD: $\text{disc}=4>0 \rightarrow 2 \text{ solutions}$.

Q32. Correct Answer: B**STEP-BY-STEP SOLUTION:**

Three fence sides: 2 perpendicular lengths L + 1 width w . $w+2L=60 \rightarrow L=(60-w)/2$. $\text{Area}=wL=w(60-w)/2=448$... Actually: label w =side parallel to barn (one fence side), L =perpendicular sides (two fence sides). Fencing: $w+2L=60 \rightarrow L=(60-w)/2$. $\text{Area}=w \times L=w(60-w)/2=448 \rightarrow w(60-w)=896$. Hmm - option B is $w(60-2w)=448$, implying fencing= $2w+L=60 \rightarrow L=60-2w \rightarrow \text{Area}=w(60-2w)=448$. This works when w is the perpendicular side (two of them) and $L=60-2w$ is the parallel side. Both setups are valid definitions of w ; option B matches the second labeling which is the standard DSAT convention.

WHY EACH OPTION IS WRONG:

- A) $w(60-w)$: implies only one w side is fenced
- C) Linear, not area equation
- D) w^2+60w has no geometric meaning

TRAP ALERT: Identifying which dimension is w . With 2 perpendicular sides: $2w+L=60 \rightarrow \text{area}=w(60-2w)$.

KEY INSIGHT: Draw the enclosure: barn=one long side; fence=2 short sides + 1 long side. Let w =short side.

FAST METHOD: $L=60-2w \rightarrow \text{area}=w(60-2w)=448$.

Q33. Correct Answer: B**STEP-BY-STEP SOLUTION:**

x -intercepts at 2 and 8. Vertex $x=(2+8)/2=5$. $k=y(5)=(5-2)(5-8)=3 \times (-3)=-9$

WHY EACH OPTION IS WRONG:

- A) $-16=y(0)$, not vertex
- C) $5=h$ (x -coord), not k
- D) 10 =sum of intercepts

TRAP ALERT: Reporting the x -coordinate ($h=5$) as the answer instead of y -coordinate ($k=-9$).

KEY INSIGHT: Vertex x =midpoint of roots. Then substitute to find y .

FAST METHOD: $h=5$; $k=(3)(-3)=-9$.

Q34. Correct Answer: A

STEP-BY-STEP SOLUTION:

$(2x-3)(x+4)=0 \rightarrow x=3/2$ or $x=-4$. Since $x>0$: $x=3/2$

WHY EACH OPTION IS WRONG:

B) $x=2$: $2(4)+5(2)-12=6 \neq 0$

C) $x=3$: $2(9)+15-12=21 \neq 0$

D) $x=4$: $2(16)+20-12=40 \neq 0$

TRAP ALERT: Stopping at $x=-4$ without checking $x>0$ condition.

KEY INSIGHT: AC method: product=-24, sum=+5 $\rightarrow$ +8 and -3. Factor to get $(2x-3)(x+4)$.

FAST METHOD: $(2x-3)(x+4)=0 \rightarrow x=3/2$ (positive).

Q35. Correct Answer: A

STEP-BY-STEP SOLUTION:

Roots -2 and 7. Sum=5=-m $\rightarrow$ m=-5. Product=-14=n. Answer: m=-5, n=-14.

WHY EACH OPTION IS WRONG:

B) m=5: sum would be -5, but $-2+7=+5$

C) n=14: product is -14, not +14

D) Both wrong

TRAP ALERT: Vieta's: sum=-m. Since sum=+5, m=-5 (not +5).

KEY INSIGHT: For x^2+mx+n : sum of roots=-m, product=n.

FAST METHOD: Sum=5=-m $\rightarrow$ m=-5; product=-14=n.

Q36. Correct Answer: C

STEP-BY-STEP SOLUTION:

Sub $y=2x-3$: $2x-3=x^2-2x-3 \rightarrow x^2-4x=0 \rightarrow x(x-4)=0 \rightarrow x=0$ or 4. Sum=4

WHY EACH OPTION IS WRONG:

A) 0 is one x-value

B) 2 is neither solution

D) 6 incorrect

TRAP ALERT: Adding y-values or not completing the substitution.

KEY INSIGHT: Set equations equal $\rightarrow$ quadratic $\rightarrow$ roots give x-values.

FAST METHOD: $x^2-4x=0 \rightarrow x=0$ or 4 $\rightarrow$ sum=4.

Q37. Correct Answer: B

STEP-BY-STEP SOLUTION:

$$3x^2 - 27 = 0 \rightarrow 3x^2 = 27 \rightarrow x^2 = 9$$

WHY EACH OPTION IS WRONG:

- A) $x^2 = 3$: $3(3) - 27 = -18 \neq 0$
- C) $(x-3)^2 = 0$ is a different equation
- D) $x(x-3) = 0$: $x = 0$ gives $3(0) - 27 = -27 \neq 0$

TRAP ALERT: Thinking $(x-3)^2 = 0$ is equivalent; it is not.

KEY INSIGHT: Simply divide both sides by 3.

FAST METHOD: $x^2 = 9$.

Q38. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$f(x) = a(x-4)^2 - 7. \quad f(0) = a(16) - 7 = 9 \rightarrow 16a = 16 \rightarrow a = 1. \quad \text{So } f(x) = (x-4)^2 - 7.$$

WHY EACH OPTION IS WRONG:

- B) $(x+4)^2$: vertex at $x = -4$, not $x = 4$
- C) Constant +9 is not the vertex k -value
- D) $a = 2$: $f(0) = 32 - 7 = 25 \neq 9$

TRAP ALERT: Using $f(0) = 9$ as the constant in vertex form instead of solving for a .

KEY INSIGHT: Write vertex form first; use the given point to find a .

FAST METHOD: $a(16) - 7 = 9 \rightarrow a = 1$.

Q39. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$q(x) = p(x+4) = (x+4)^2 - (x+4) - 6 = x^2 + 7x + 6 = (x+1)(x+6) = 0 \rightarrow x = -1 \text{ or } x = -6. \quad \text{Since } x < -2: x = -6$$

WHY EACH OPTION IS WRONG:

- B) -5 is not a root of q
- C) -2 is a root of $p(x)$ before the shift
- D) -1 is a root of q but fails $x < -2$

TRAP ALERT: Using roots of p directly instead of computing q 's roots via the shift.

KEY INSIGHT: Shift: roots of p are 3 and -2. Shift left 4: roots of q are -1 and -6.

FAST METHOD: Roots: -1 and -6; $x < -2 \rightarrow x = -6$.

Q40. Correct Answer: 3

STEP-BY-STEP SOLUTION:

$$\text{disc}=(-6)^2-4(3)(k)=36-12k=0 \rightarrow k=3$$

TRAP ALERT: Setting $\text{disc}>0$ or <0 instead of $=0$.

KEY INSIGHT: Exactly one solution requires $\text{disc}=0$.

FAST METHOD: $36=12k \rightarrow k=3$.

Q41. Correct Answer: B

STEP-BY-STEP SOLUTION:

Shortcut: $a+b+c=f(1)$. Since $x=1$ is a root: $f(1)=0$. Therefore $a+b+c=0$.

WHY EACH OPTION IS WRONG:

A) -5: $f(0)=c=5$, not the sum

C) 1: $f(1)=0$ equals $a+b+c$

D) 5= c alone

TRAP ALERT: Computing a, b, c individually when $f(1)=a+b+c$ is the shortcut.

KEY INSIGHT: $a+b+c=f(1)$. Since $(1,0)$ is on the graph, $f(1)=0$.

FAST METHOD: $f(1)=a+b+c=0$ because $x=1$ is a root.

Q42. Correct Answer: B

STEP-BY-STEP SOLUTION:

$$y=x^2-x-12=(x-4)(x+3). \text{ x-intercepts at } x=4 \text{ and } x=-3.$$

A: $(x+4)(x-3)$: intercepts at -4 and 3 (WRONG)

B: $-2(x-4)(x+3)$: same factors $\rightarrow$ same intercepts ✓

WHY EACH OPTION IS WRONG:

A) $(x+4)(x-3)$: intercepts at $x=-4$ and $x=3$ (different)

C) $3(x+4)(x-3)$: intercepts at -4 and 3 (wrong)

D) $-(x+4)(x-3)$: intercepts at -4 and 3 (wrong)

TRAP ALERT: Vertical scaling (multiplying by -2) does NOT change x-intercepts.

KEY INSIGHT: Same x-intercepts means same factors $(x-r)$. Multiplying by any nonzero constant preserves zeros.

FAST METHOD: Only B uses factors $(x-4)(x+3)$.

Q43. Correct Answer: D

STEP-BY-STEP SOLUTION:

$$l(l-7)=120 \rightarrow l^2-7l-120=0 \rightarrow (l-15)(l+8)=0 \rightarrow l=15$$

WHY EACH OPTION IS WRONG:

- A) 8 is the WIDTH ($15-7=8$)
- B) $10 \times 3 = 30 \neq 120$
- C) $12 \times 5 = 60 \neq 120$

TRAP ALERT: Finding $l=15$ but reporting width=8.

KEY INSIGHT: Identify what you solved for vs. what was asked.

FAST METHOD: $(l-15)(l+8)=0 \rightarrow l=15$.

Q44. Correct Answer: B

STEP-BY-STEP SOLUTION:

$$\text{Sub } x=-2 \text{ for any } k: (-2)^2 + (k+2)(-2) + 2k = 4 - 2k - 4 + 2k = 0 \checkmark \text{ for ALL } k$$

WHY EACH OPTION IS WRONG:

- A) $x=-k$ also works but varies with k
- C) $x=k$: $2k^2 + 4k \neq 0$ generally
- D) $x=2$: $8 + 4k \neq 0$ generally

TRAP ALERT: Trying only one k value and concluding an option always works.

KEY INSIGHT: $x=-2$ makes the expression identically 0 for every value of k .

FAST METHOD: Sub $x=-2$: k terms cancel $\rightarrow$ always 0.

Q45. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$(x-3)^2=25 \rightarrow x-3=\pm 5 \rightarrow x=8 \text{ or } x=-2. \text{ Product}=8 \times (-2)=-16$$

WHY EACH OPTION IS WRONG:

- B) -8: confusing sum and product
- C) 8: one root, not product
- D) 16: absolute value of product

TRAP ALERT: Computing sum instead of product.

KEY INSIGHT: Expand: $x^2-6x-16=0$. By Vieta's: product= $c/a=-16$.

FAST METHOD: $x=8$ and $x=-2 \rightarrow$ product=-16.

Q46. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$g(x)=f(x+3)=(x+3)^2-8(x+3)+12=x^2+6x+9-8x-24+12=x^2-2x-3$$

WHY EACH OPTION IS WRONG:

- B) x^2+2x+3 : sign errors
- C) $x^2-14x+39$: computed $f(x-3)$ not $f(x+3)$
- D) x^2-2x+3 : constant term wrong: $9-24+12=-3$

TRAP ALERT: Error in $-8(x+3)=-8x-24$ (not $-8x+24$).

KEY INSIGHT: Replace every x in f with $(x+3)$, expand carefully.

FAST METHOD: $(x+3)^2=x^2+6x+9$; $-8(x+3)=-8x-24$; $\text{sum}=x^2-2x-3$.

Q47. Correct Answer: C

STEP-BY-STEP SOLUTION:

$$\text{Sub } y=3-2x: 3-2x=x^2-10x+c \rightarrow x^2-8x+(c-3)=0. \text{ disc}=0: 64-4(c-3)=0 \rightarrow 76=4c \rightarrow c=19$$

WHY EACH OPTION IS WRONG:

- A) $c=15$: $\text{disc}=64-48=16>0$ (two solutions)
- B) $c=17$: $\text{disc}=64-56=8>0$ (two solutions)
- D) $c=22$: $\text{disc}=64-76=-12<0$ (no solutions)

TRAP ALERT: Not setting discriminant equal to zero.

KEY INSIGHT: Substitute line into parabola; set discriminant=0 for one solution.

FAST METHOD: $64-4(c-3)=0 \rightarrow c=19$.

Q48. Correct Answer: D

STEP-BY-STEP SOLUTION:

$$\text{disc}=0: 16-4k(k-3)=0 \rightarrow k^2-3k-4=0 \rightarrow (k-4)(k+1)=0 \rightarrow k=4 \text{ or } k=-1. \text{ Answer D: } k=4.$$

WHY EACH OPTION IS WRONG:

- A) $k=-4$: $\text{disc}=16-112<0$
- B) $k=-1$ also works (other valid answer)
- C) $k=1$: $\text{disc}=16+8=24\neq 0$

TRAP ALERT: Choosing $k=1$ by guessing without computing discriminant.

KEY INSIGHT: Two equal roots: $\text{disc}=0$. Solve resulting quadratic for k .

FAST METHOD: $16-4k(k-3)=0 \rightarrow k=4 \text{ or } -1$.

Q49. Correct Answer: A

STEP-BY-STEP SOLUTION:

Complete the square: $2(x^2-6x)+19=2(x-3)^2-18+19=2(x-3)^2+1$. Minimum=1.

WHY EACH OPTION IS WRONG:

- B) $2(x-3)^2+7$: arithmetic error, $-18+19=+1$ not $+7$
- C) $2(x-6)^2+1$: axis is $x=3$, not $x=6$
- D) Doesn't display minimum as explicit constant

TRAP ALERT: Arithmetic: $-2(9)+19=-18+19=+1$ (not $+7$).

KEY INSIGHT: Vertex form $a(x-h)^2+k$ shows minimum value k (for $a>0$).

FAST METHOD: $h=3$; $k=-18+19=1$.

Q50. Correct Answer: C

STEP-BY-STEP SOLUTION:

$\text{disc}=0 \rightarrow x=(-b \pm 0)/(2a)=-b/(2a)$. Exactly one real solution equal to $-b/(2a)$.

WHY EACH OPTION IS WRONG:

- A) No solutions: $\text{disc}<0$
- B) Two distinct: $\text{disc}>0$
- D) Option D says $b/(2a)$ — missing the negative sign (a trap)

TRAP ALERT: Option D is a deliberate trap: the formula is $-b/(2a)$, NOT $+b/(2a)$.

KEY INSIGHT: One repeated root = parabola tangent to x-axis. Root= $-b/(2a)$.

FAST METHOD: $\text{disc}=0 \rightarrow x=-b/(2a)$ exactly once.

Q51. Correct Answer: C

STEP-BY-STEP SOLUTION:

$x=(4 \pm \sqrt{16-4})/2=(4 \pm \sqrt{12})/2=(4 \pm 2\sqrt{3})/2=2 \pm \sqrt{3}$. $p=2$, $q=1$. $p+q=3$

WHY EACH OPTION IS WRONG:

- A) $p+q=1$: $x=\sqrt{3}$ is not a root
- B) $p+q=2$: no radical term then
- D) $p+q=4$: coefficient of $\sqrt{3}$ would be 2, giving $2 \pm 2\sqrt{3}$ which doesn't satisfy equation

TRAP ALERT: $\sqrt{12}=2\sqrt{3}$, NOT $2\sqrt{6}$.

KEY INSIGHT: Simplify: $\sqrt{12}=\sqrt{4 \times 3}=2\sqrt{3}$. Read p and q carefully.

FAST METHOD: $x=2 \pm \sqrt{3} \rightarrow p=2$, $q=1 \rightarrow \text{sum}=3$.

Q52. Correct Answer: A**STEP-BY-STEP SOLUTION:**

Table shows minimum at (2,-3). Vertex form: $(x-2)^2-3$.

Verify: $f(-1)=9-3=6$ ✓; $f(0)=4-3=1$ ✓; $f(1)=1-3=-2$ ✓

WHY EACH OPTION IS WRONG:

B) $-(x-2)^2-3$: opens downward; $f(0)=-4-3=-7 \neq 1$

C) $(x+2)^2-3$: vertex at $x=-2$; $f(-1)=1-3=-2 \neq 6$

D) $(x+2)^2+3$: $f(0)=4+3=7 \neq 1$

TRAP ALERT: Options B and C partially match; always verify multiple points.

KEY INSIGHT: Identify vertex from table (minimum), write vertex form, verify.

FAST METHOD: Vertex (2,-3) $\rightarrow (x-2)^2-3$. Check $f(0)=1$ ✓

Q53. Correct Answer: B**STEP-BY-STEP SOLUTION:**

Set $h=0$: $-16t^2+24t+50=0$. Multiply by -1: $16t^2-24t-50=0$. $8t^2-12t-25=0$.

Quadratic formula: $t=(12+\sqrt{144+800})/16=(12+\sqrt{944})/16=(12+30.72)/16=42.72/16 \approx 2.67$.

Nearest tenth = 2.7. (Reject the negative root.)

WHY EACH OPTION IS WRONG:

A) $t=2.3$: $h(2.3)=-16(5.29)+24(2.3)+45=15.6 \neq 0$

C) $t=3.1$: past ground level

D) $t=3.5$: far past

TRAP ALERT: Using the negative root of the quadratic formula.

KEY INSIGHT: Reject negative t . Use the positive root only.

FAST METHOD: $t=(24+\sqrt{3456})/32 \approx 2.59$, closest to B=2.7.

Q54. Correct Answer: 2**STEP-BY-STEP SOLUTION:**

$y=x+1$ from second equation. Sub: $x^2+(x+1)=7 \rightarrow x^2+x-6=0 \rightarrow (x+3)(x-2)=0 \rightarrow x=-3$ or $x=2$. $x>0 \rightarrow x=2$

TRAP ALERT: Solving for y first and confusing which variable is requested.

KEY INSIGHT: Substitute linear equation into quadratic, solve, apply condition.

FAST METHOD: $x^2+x-6=0 \rightarrow x=2$.

Q55. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$f(x)=(x+2)(x-5)+10=x^2-3x-10+10=x^2-3x=x(x-3)$. So $g(x)=(x+a)(x+b)$ matches $x(x-3)$. Thus $a=0$, $b=-3$ (or $a=-3$, $b=0$). $|a-b|=|0-(-3)|=3$

WHY EACH OPTION IS WRONG:

- A) 1: wrong
- C) 5: used $(x+2)(x-5)$ before adding 10
- D) 7: $|2-(-5)|=7$ ignores the +10 adjustment

TRAP ALERT: Using factors $(x+2)(x-5)$ of the original before adding 10.

KEY INSIGHT: Expand f fully: +10 cancels the -10, simplifying everything.

FAST METHOD: $f(x)=x^2-3x=x(x-3) \rightarrow a=0, b=-3 \rightarrow |a-b|=3$.

Q56. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$y=x^2-6x+5=(x-3)^2-4$. Vertex $(3,-4)$.

Shift left 2: vertex $\rightarrow (1,-4)$.

Shift up 4: vertex $\rightarrow (1,0)$.

Minimum=0

WHY EACH OPTION IS WRONG:

- A) -4: original minimum before shifts
- C) 4: amount shifted up
- D) 8: incorrect

TRAP ALERT: Applying shifts to wrong component of vertex.

KEY INSIGHT: Shift left: h decreases. Shift up: k increases. New $k=-4+4=0$.

FAST METHOD: New $k=-4+4=0$.

Q57. Correct Answer: A**STEP-BY-STEP SOLUTION:**

$r-s=8$, $r+s=4 \rightarrow r=6$, $s=-2$. Product $q=rs=6 \times (-2)=-12$

WHY EACH OPTION IS WRONG:

- B) -20: incorrect
- C) -24: would need roots like 6 and -4
- D) -32: impossible for these roots

TRAP ALERT: Using sum as product or vice versa.

KEY INSIGHT: Product of roots = q (for monic quadratic x^2+px+q).

FAST METHOD: $r=6$, $s=-2 \rightarrow q=-12 \rightarrow$ answer A.

Q58. Correct Answer: -8

STEP-BY-STEP SOLUTION:

Vertex at $x=4$: $-b/(2 \times 1)=4 \rightarrow b=-8$.

Alternative: root at $x=6$, axis at $x=4 \rightarrow$ other root at $x=2$. $h(x)=(x-6)(x-2)=x^2-8x+12 \rightarrow b=-8$.

TRAP ALERT: Using single root without finding axis of symmetry.

KEY INSIGHT: Axis of symmetry $= -b/2a$. For monic: $-b/2=4 \rightarrow b=-8$.

FAST METHOD: $b=-2 \times 1 \times 4=-8$.

Q59. Correct Answer: A

STEP-BY-STEP SOLUTION:

$g(x)=(x-k)^2-4=0 \rightarrow (x-k)^2=4 \rightarrow x-k=\pm 2 \rightarrow x=k\pm 2$

WHY EACH OPTION IS WRONG:

B) ± 4 : $\sqrt{4}=2$ not 4

C) $-k\pm 2$: wrong sign on k

D) $k^2\pm 4$: no connection to factored form

TRAP ALERT: Taking $\sqrt{4}=4$ instead of 2.

KEY INSIGHT: $(x-k)^2=4 \rightarrow x-k=\pm 2$.

FAST METHOD: $x=k\pm 2$.

Q60. Correct Answer: 2

STEP-BY-STEP SOLUTION:

Set equal: $2x^2-8x+3=3-4x \rightarrow 2x^2-4x=0 \rightarrow 2x(x-2)=0 \rightarrow x=0$ or $x=2$. Positive $=2$

TRAP ALERT: Arithmetic error: $-8x+4x=-4x$ (not $-12x$).

KEY INSIGHT: Collect all terms on one side, factor.

FAST METHOD: $2x^2-4x=0 \rightarrow x=2$.

TOPIC SUMMARY — Quadratic Equations - Solving

Common Mistakes:

1. Not rearranging to standard form $ax^2+bx+c=0$ before solving.
2. Vieta's sign trap: sum of roots $= -b/a$ (NOT $+b/a$). Since $\text{sum}=-b$, if roots sum to S , then $b=-S$.
3. Finding all roots but not applying the stated condition ($x>0$, $x>2$, etc.) to select the correct one.

Must Remember:

1. Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

2. Discriminant: >0 gives two distinct roots; $=0$ gives one repeated root; <0 gives no real roots
3. Vertex form $y = a(x-h)^2 + k$: vertex at (h,k) ; minimum/maximum value is k

Expected on DSAT:

2-4 quadratic questions per Digital SAT (1-2 in Module 1, 1-2 in Module 2). Hard module often tests completing the square, discriminant conditions, or function transformations.
